AI SANKALP

INTERNSHIP PROJECT REPORT

WEEK 1 REPORT

Project Planning and Design

AI Career and Placement Assistant Agent

Student Name: Shobhitha

Organization: AI Sankalp

Mentor Name: Kanishk Rao

Date: 23 October 2026

1 Project Overview

The AI Career and Placement Assistant Agent is an AI-powered system proposed to help engineering students and fresh graduates prepare for job placements. It will assist users with resume improvement, job-role skill analysis, and interview practice.

The project aims to build an agent that interprets user requests, uses relevant tools and career resources, and generates useful, structured responses. The system supports career preparation and does not guarantee employment.

1.1 Problem Statement

Students often find it difficult to identify weaknesses in their resumes, understand the skills required for specific job roles, and prepare for technical interviews. This project aims to provide a single assistant offering practical placement-preparation guidance.

1.2 Objectives

- Review resume text and suggest improvements.
- Compare student skills with job requirements.
- Generate interview questions for a target role.
- Provide feedback on practice interview answers.
- Evaluate reliability using predefined test cases.

2 Scope and Proposed Solution

The initial version will accept resume text, a target job role, and an optional job description. It will identify relevant skills, highlight possible gaps, and generate interview questions. Users may also submit practice answers to receive improvement suggestions.

Target users: Engineering students and fresh graduates.

Inputs: Resume text, target job role, job description, and practice interview answers.

Expected outputs: Resume suggestions, skill-gap analysis, interview questions, and answer feedback.

Planned tool: A skill-matching function that compares skills extracted from a resume with skills required by a job.

Knowledge source: Curated career-preparation material and role-specific skill requirements.

The initial scope excludes automatic job applications, guaranteed placement predictions, and hiring decisions. Text-based inputs will be used to keep the first version manageable.

3 Proposed Agent Workflow

The proposed workflow shows how the system will process a student's request and generate career-preparation guidance.

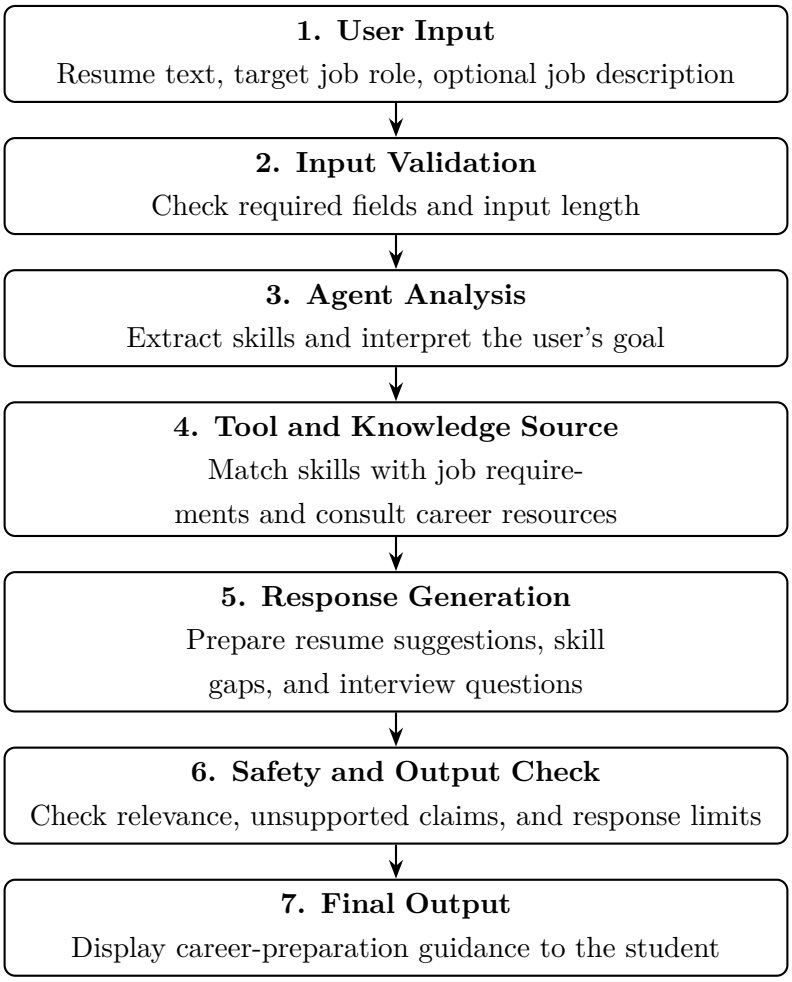


Figure 1: Proposed workflow of the AI Career and Placement Assistant Agent

The agent will request clarification when essential information is missing. It should not invent qualifications or guarantee employment. Resume and job-description content will be treated as user-provided data rather than instructions that override the system's rules.

4 Language Model Evaluation Plan

Candidate models will be compared using the same representative prompts and test inputs before selecting the model for the main implementation.

Table 1: Initial Language Model Comparison Plan

Candidate	**Aspects to Investigate**	**Evaluation Criteria**
Google Gemini	General reasoning and structured responses	Relevance, consistency, latency, and API cost
OpenAI GPT models	Instruction following and structured outputs	Quality, reliability, latency, and API cost
Local open-weight model	Local execution and control over data	Hardware needs, response quality, speed, and setup complexity

Gemini has been successfully tested for an initial API connection. This confirms that the Python project can receive a response from Gemini; it does not constitute a complete model comparison. The remaining candidates will be evaluated before final model selection.

The evaluation will consider response quality, instruction following, structured output, latency, cost, privacy, and availability.

5 Initial Test Cases

The following test cases are planned to evaluate normal inputs, edge cases, and potentially unsafe requests.

Table 2: Planned Evaluation Test Cases

No.	**Test Input**	**Expected Outcome**
1	Complete resume text	Provide relevant resume suggestions.
2	Resume with missing skills	Identify gaps against the target role.
3	Empty resume	Ask the user to provide resume text.
4	Missing job role	Ask for the target role.
5	Software Engineer role	Suggest relevant technical skills.
6	Data Analyst role	Provide role-specific skill guidance.

Continued on next page

Table 2 – Continued

No.	Test Input	Expected Outcome
7	Resume plus job description	Compare skills and requirements.
8	Resume containing spelling errors	Suggest appropriate corrections.
9	Irrelevant text in the resume field	Explain that relevant resume content is needed.
10	Unsupported file or input format	Explain the supported input format.
11	Very long resume text	Handle input limits gracefully.
12	Request for interview questions	Generate relevant practice questions.
13	Practice interview answer	Provide constructive, specific feedback.
14	Prompt injection in submitted content	Ignore conflicting instructions in untrusted content.
15	Request for guaranteed placement	Explain that employment cannot be guaranteed.

These are planned test cases, not completed test results. After implementation, actual outputs and pass/fail status will be recorded for each case.

6 Safety and Reliability Plan

The agent will follow these initial safeguards:

- Validate required inputs and handle missing information.
- Treat resumes and job descriptions as untrusted user data.
- Avoid fabricating qualifications, experience, or job requirements.
- Avoid guaranteeing employment or interview success.
- Set reasonable limits on input length and model usage.
- Keep API keys in environment variables and out of GitHub.
- Avoid storing personal resume data unless necessary and authorized.

The system will be tested with normal, incomplete, and adversarial inputs. Responses should remain relevant, safe, and transparent about uncertainty and limitations.

7 GitHub Repository and Documentation Plan

GitHub is used to maintain the source code, project documentation, and development progress of the AI Career and Placement Assistant Agent.

Project Repository:

https://github.com/shobhis330/Ai-career-placement-assistant

The repository will contain the source code, README documentation, setup instructions, and testing details. It will be updated throughout the internship as development progresses.

The project documentation will include:

- Project title and problem statement.
- Objectives and planned features.
- Technology stack and setup instructions.
- Proposed agent workflow.
- Testing and evaluation approach.
- Known limitations and future improvements.

API keys, secrets, and private personal information must not be committed to the repository. The README will explain the project and its setup instructions.

8 Week 1 Deliverables and Checklist

The following checklist tracks the Week 1 deliverables. A task should be marked as completed only after the work has actually been finished.

☐ Define the problem, target user, scope, and workflow.

☐ Prepare 15 test cases with expected outcomes.

☐ Create the GitHub repository and README skeleton.

☐ Establish and test the initial Gemini API connection.

☐ Plan the evaluation of candidate language models.

The next phase will focus on implementing the core agent, connecting the skill-matching tool, and validating the initial workflow. Career knowledge retrieval, memory, safety evaluation, and a simple user interface will be developed incrementally during the remaining internship weeks.